

模拟电子技术基础

Analog Device

4 Integrated operational circuit (IOC)

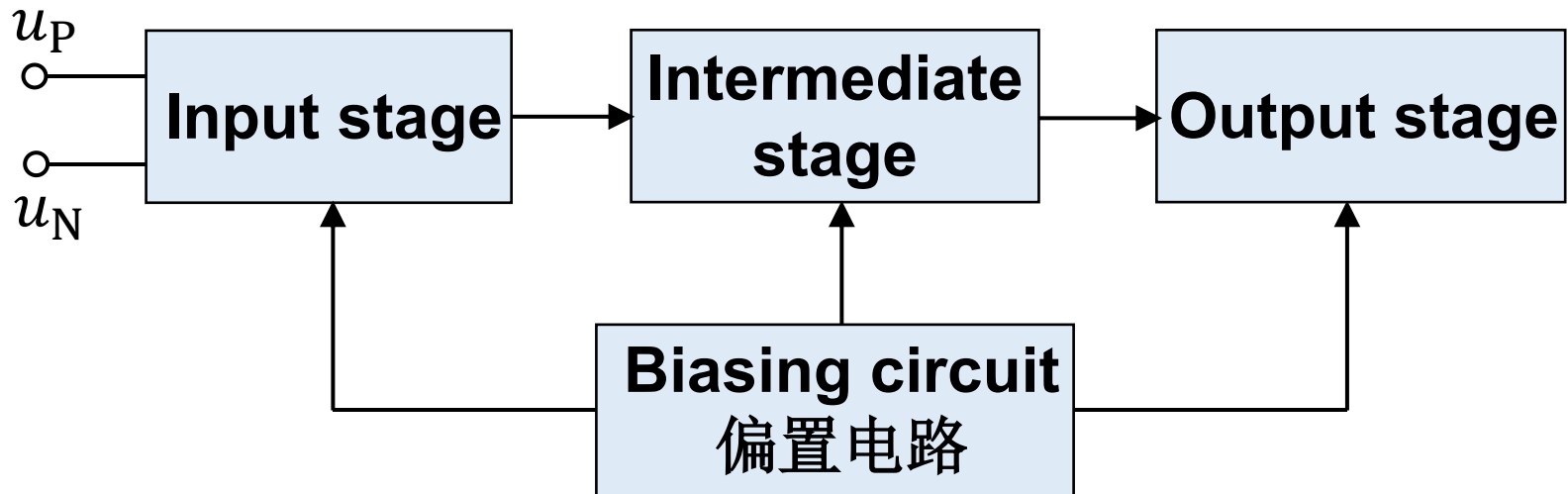
集成运放电路

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4.0 Structure of IOC

The structure of the integrated operational circuit (IOC)



Intermediate stage中间级: high voltage gain

Use **common-e amplifier circuit**共射极放大电路
(composite amplifier复合管, Multistage amplifier多级放大电路)

4.1 Multistage amplifier 多级放大电路

□ Typical specification for a general purpose operational amplifier

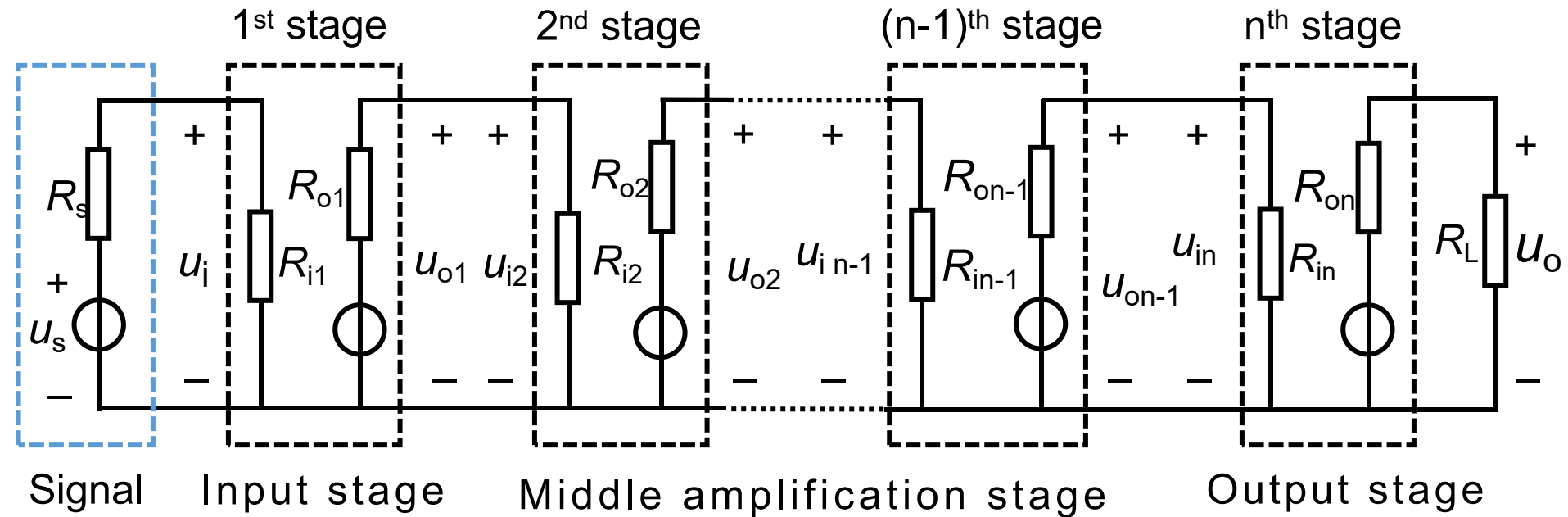
- 1) Input resistance $\sim 1\text{M}\Omega$
- 2) Output resistance $\sim 100\ \Omega$
- 3) Voltage gain $\sim 100,000$

□ No single transistor amplifier can satisfy the specification

□ Cascading multiple stages of amplifier to meet the spec
多级串联放大器可以满足要求

4.1.1 Multistage amplifier circuit

Equivalent circuit



4.1.1 The coupling mode of multistage amplifier circuit

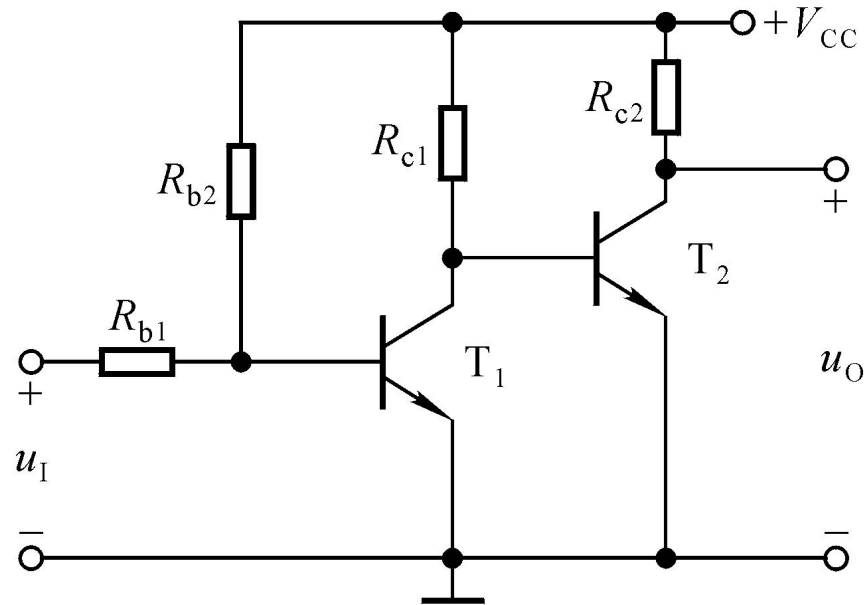
□ Coupling mode 耦合方式

1) Direct coupling 直接耦合

2) Resistance-capacitance coupling 阻容耦合

3) Transformer coupling 变压器耦合

1) Direct coupling circuit 直接耦合电路



□ Merits of direct coupling circuit:

Can amplify both AC and DC signal

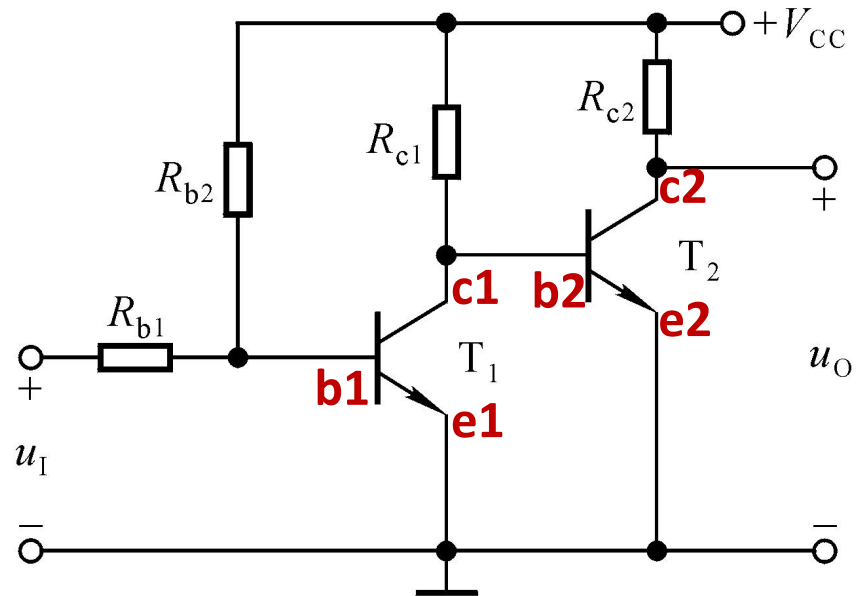
No need of capacitor and inductor, easy to integrate

□ Drawback of direct coupling circuit:

a) The quiescent working points of T_1 and T_2 are dependent, inconvenient to design and diagnose the circuit

$$U_{BEQ2} = U_{CEQ1}$$

b) U_{CEQ1} too small, T_1 is close to saturation region

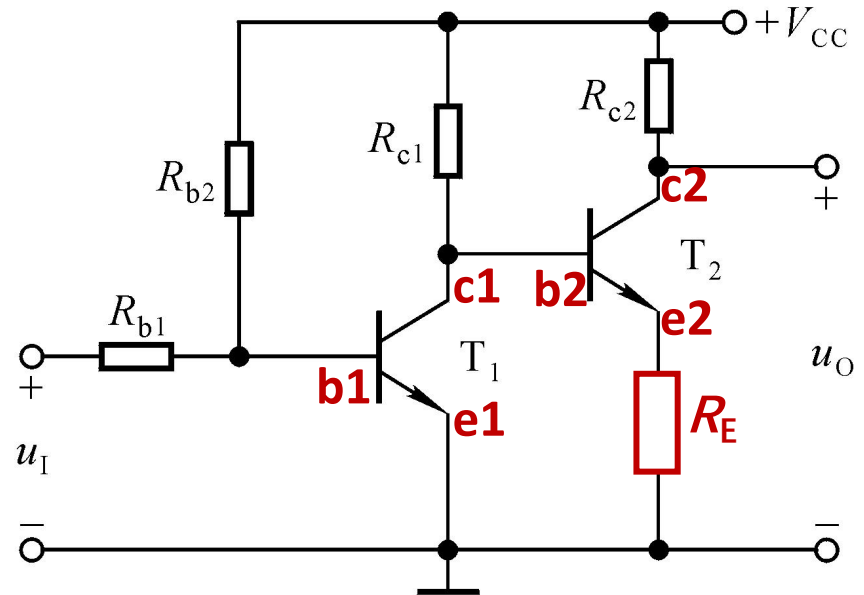


How to find a proper quiescent working point of T1?

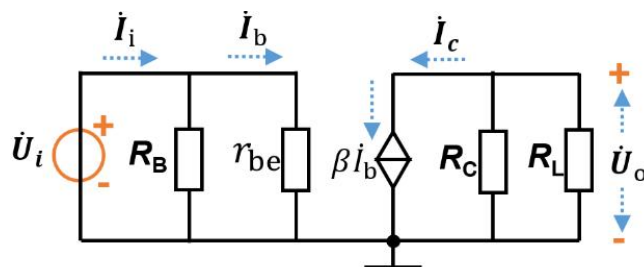
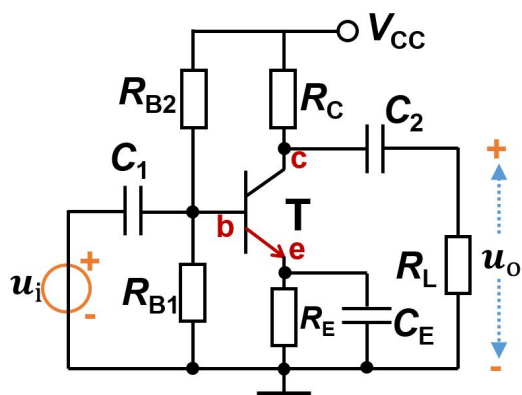
Raise $U_{C1} \rightarrow U_{B2} \rightarrow U_{E2}$!

Introduce resistor R_E

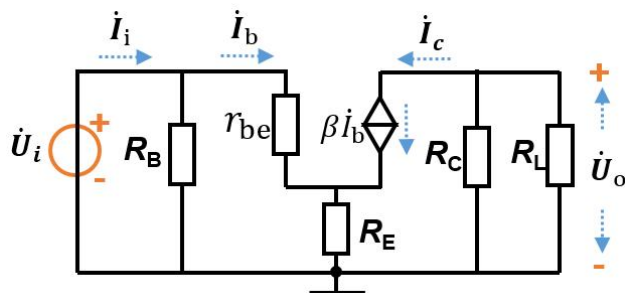
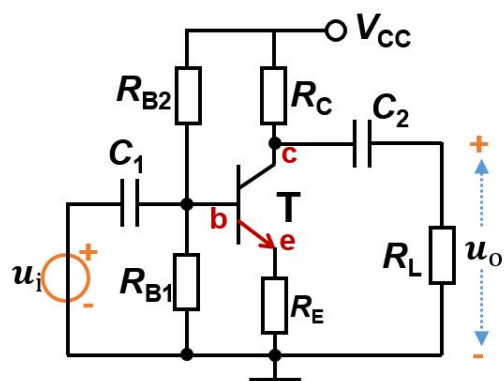
$$U_{CEQ1} = U_{B2} = U_{BEQ2} + I_{E2} R_E$$



A new problem will be introduced:



$$\dot{A}_u = -\frac{\beta R_L'}{r_{be}} = -\frac{\beta(R_L || R_C)}{r_{be}} \approx -126$$



$$\dot{A}_u = -\frac{\beta(R_C || R_L)}{r_{be} + (1 + \beta)R_E} = -1.72$$

After introducing resistor R_E , although we increase the U_{CEQ1} , the voltage gain of T_2 will decrease significantly!

The idea case is: we have a high U_{E2} and small R_E

1) If we need $U_{CEQ1} < 2\text{ V}$

Use PN diode to replace R_E

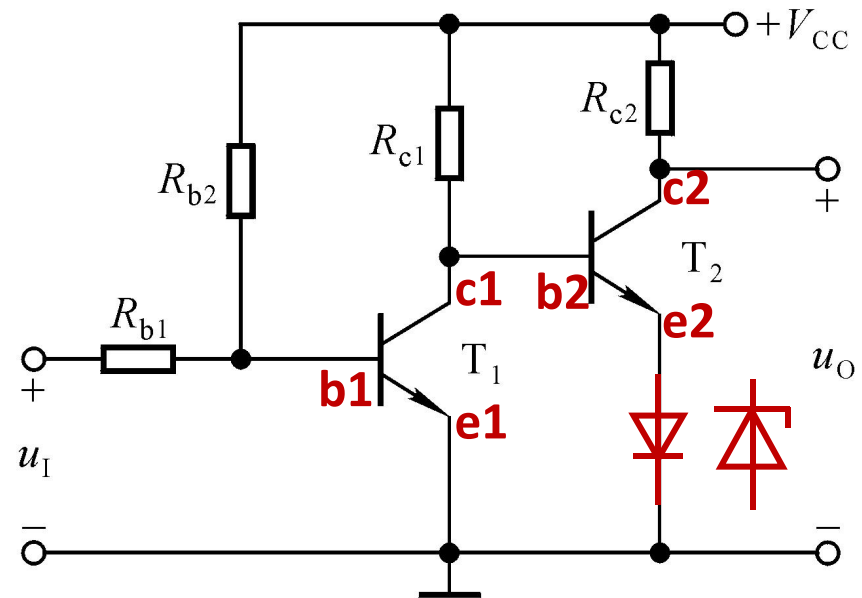
$$U_{E2} = U_{ON} + I_{E2} r_D$$

2) If we need high $U_{CEQ1} > 2\text{ V}$

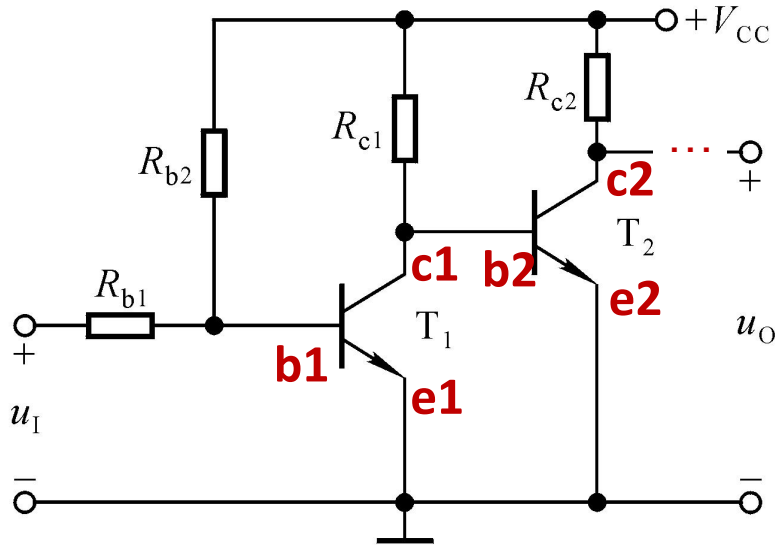
Use voltage regulator diode to replace R_E

$$U_{E2} = U_Z + I_{E2} r_Z$$

$$U_Z > U_{ON} \text{ and } r_Z < r_D$$



If there are multiple stages:



$$U_{CQ1} (U_{BQ2}) > U_{BQ1}$$

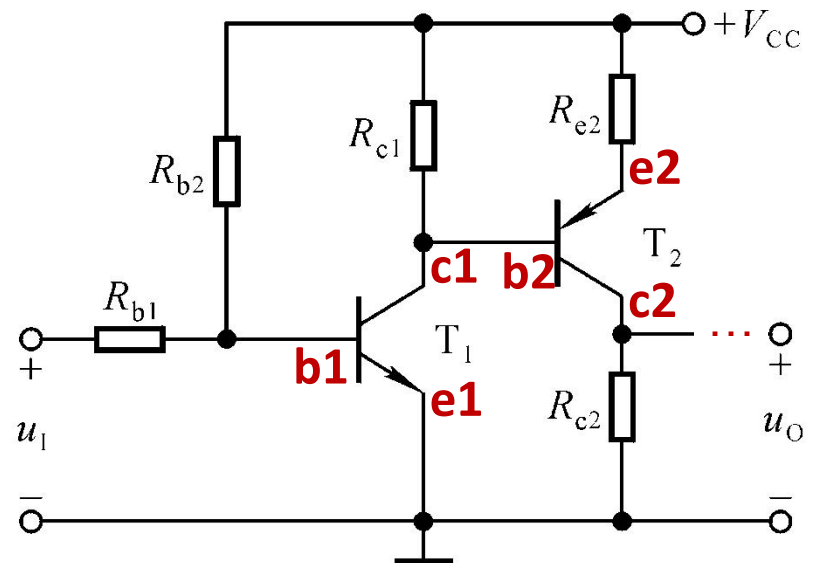
$$U_{CQ2} > U_{CQ1}$$

$$U_{CQN} > U_{CQN-1} \dots > U_{CQ1}$$

Problem: V_{CC} will be very large

Solutions:

T1: NPN, T2: PNP, T3: NPN, T4: PNP...



$$U_{CQ1} (U_{BQ2}) > U_{BQ1}$$

$$U_{CQ2} < U_{CQ1}$$

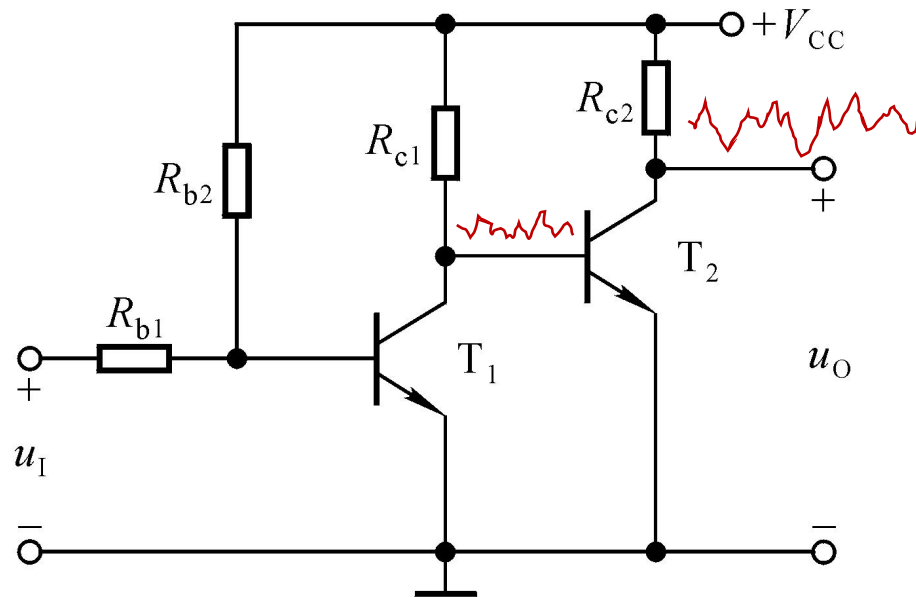
$$U_{CQ3} > U_{CQ2}, U_{CQ4} < U_{CQ3} \dots$$

□ Drawback of direct coupling circuit:

c) Zero-drift error 零点漂移

When input signal $u_i=0$, the output signal u_o is not stable, but shows a slow and random variation

Zero-drift error is due to the drift of quiescent working point



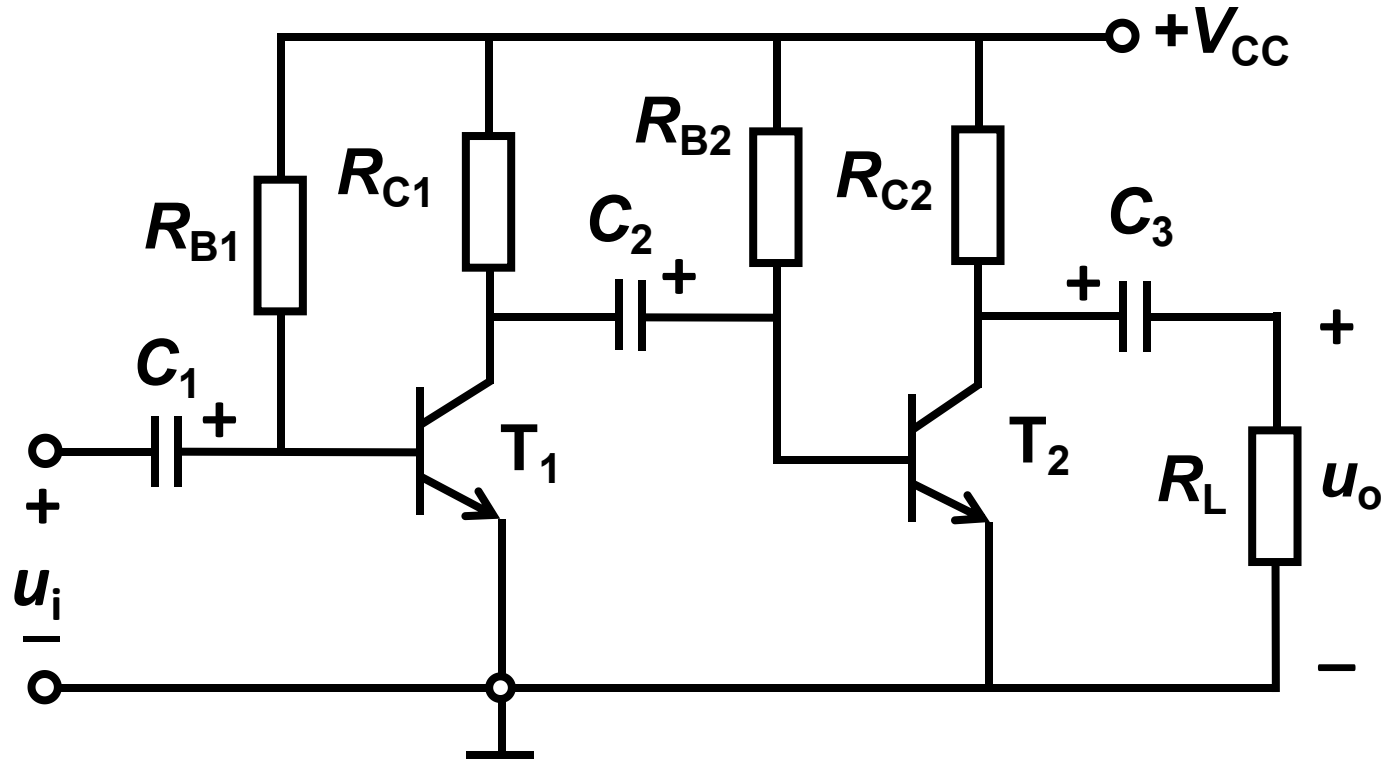
Reasons that contribute to the zero-drift error:

- a) Temperature 由温度引起的零漂称为温漂
- b) Aging 由元器件老化引起的零漂称为时漂

Suppress the zero-drift error:

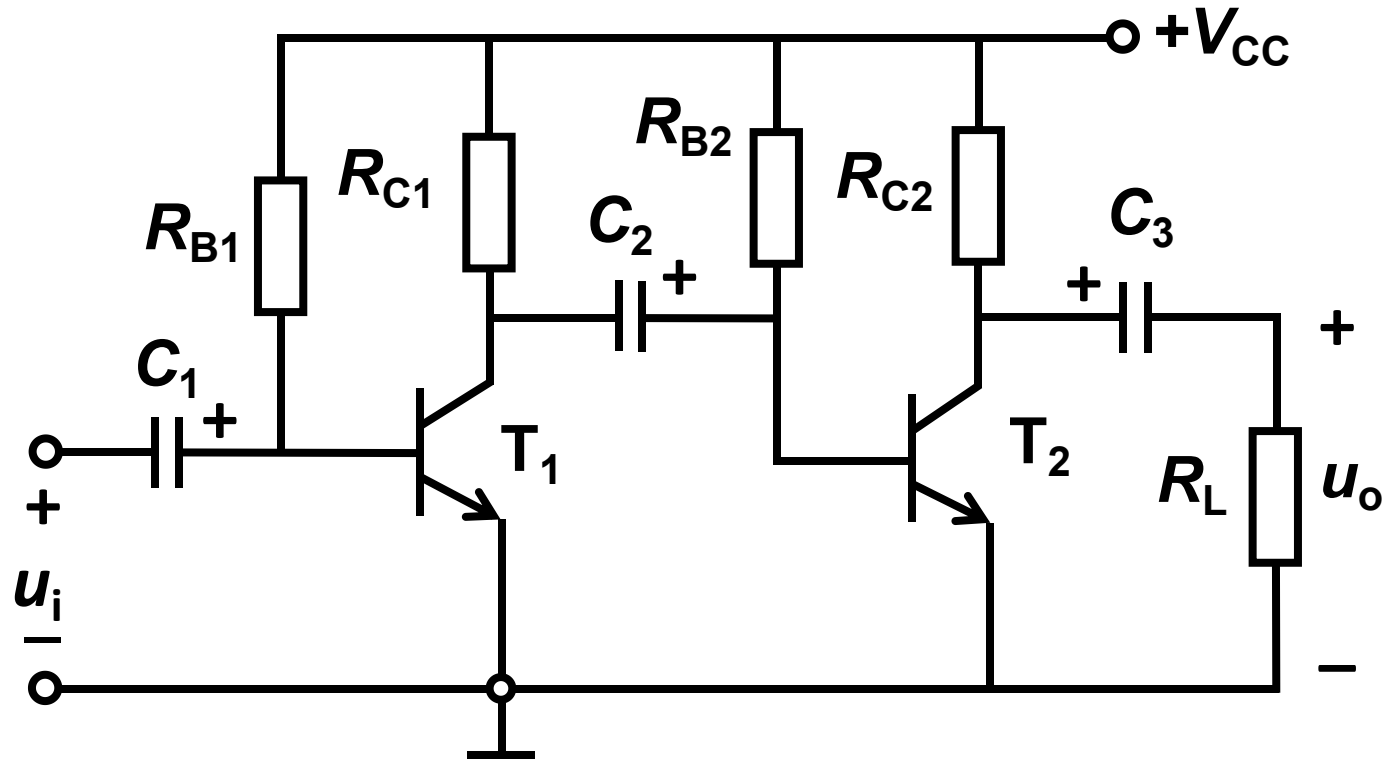
- a) Negative feedback DC circuit 直流负反馈电路
- b) Differential amplifier circuit 采用差分放大电路

2) Resistance-capacitance coupling amplification circuit 阻容耦合放大电路



□ Merits of RC coupling circuit:

The quiescent working points of T_1 and T_2 are independent

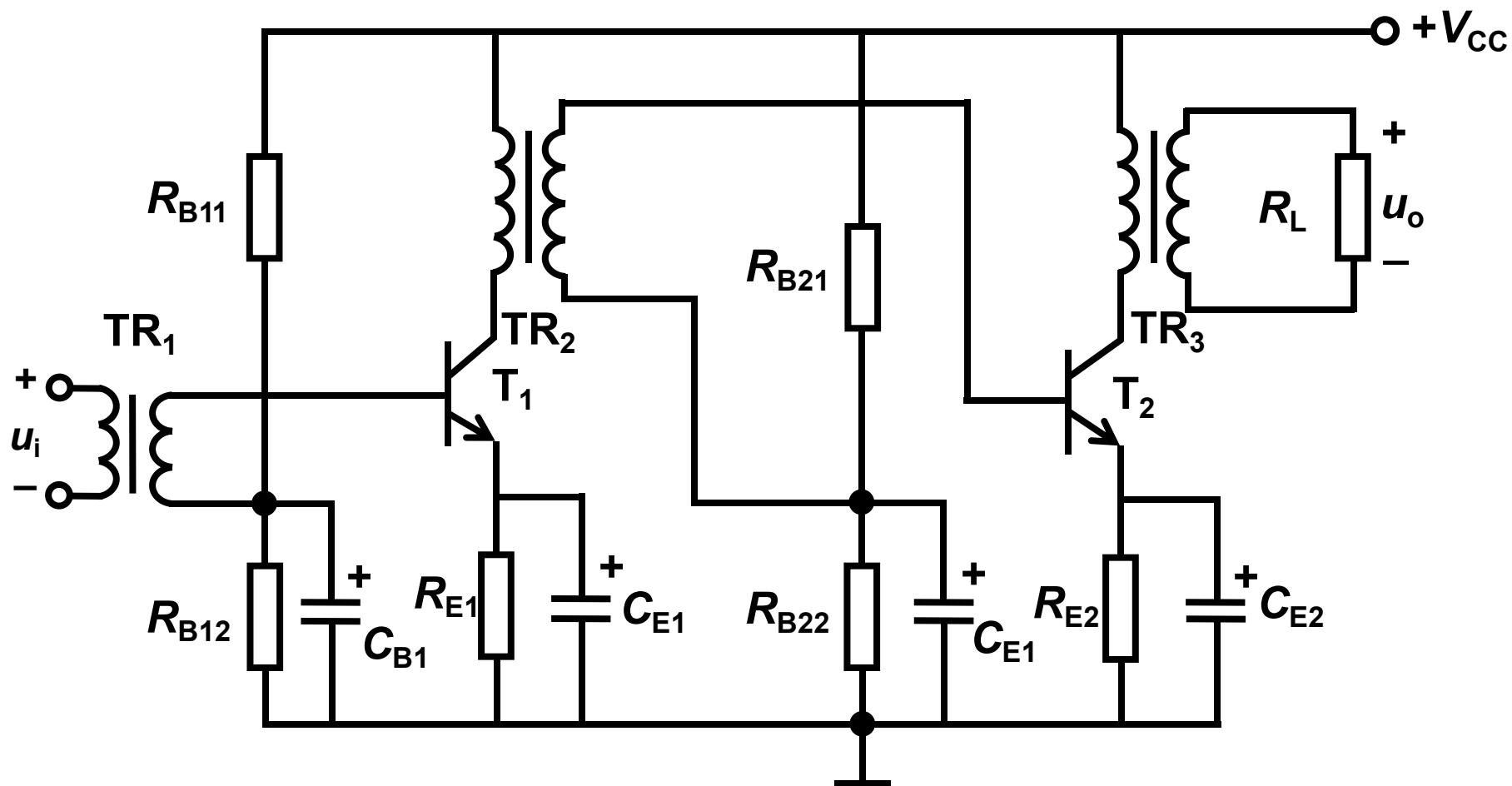


□ Drawback of RC coupling circuit:

Cannot amplify DC and low frequency signal

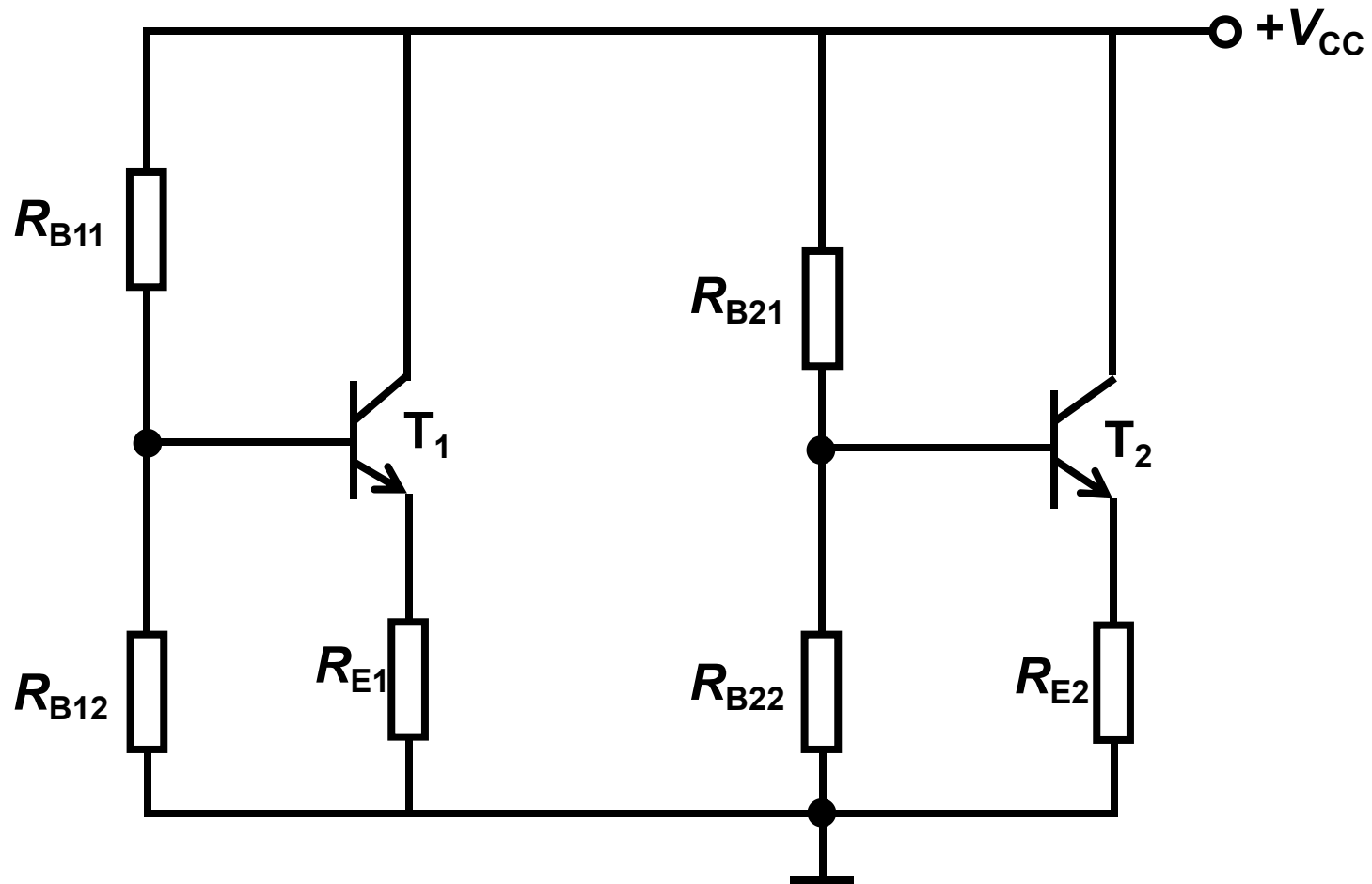
Cannot be applied to integrated circuits, due to high space consumption of capacitor with large capacitance

3) Transformer coupling amplification circuit 变压耦合放大电路

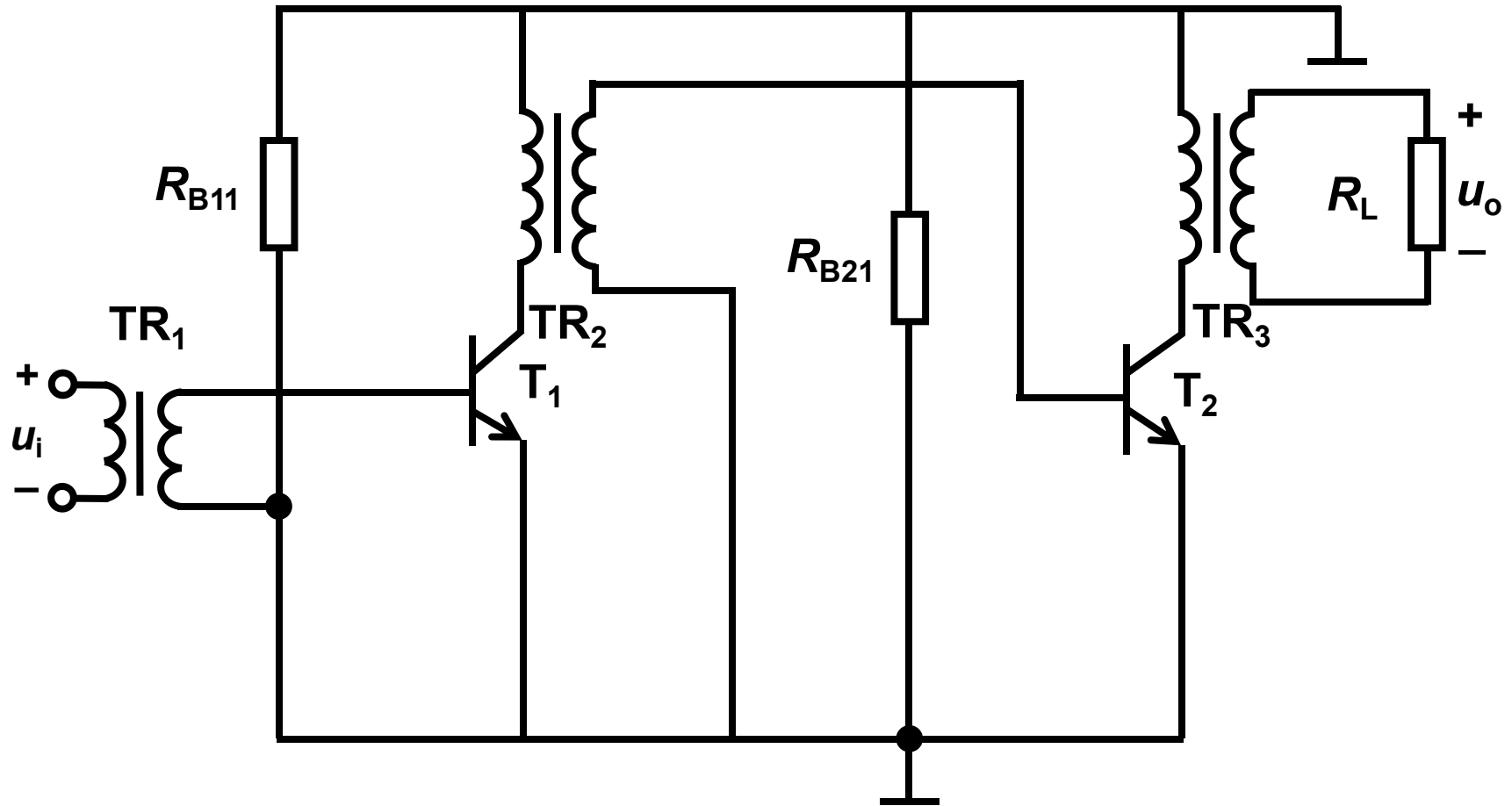


DC circuit

Transformer and inductor is short for DC circuits 变压器和电感对直流不响应，可以直接短路

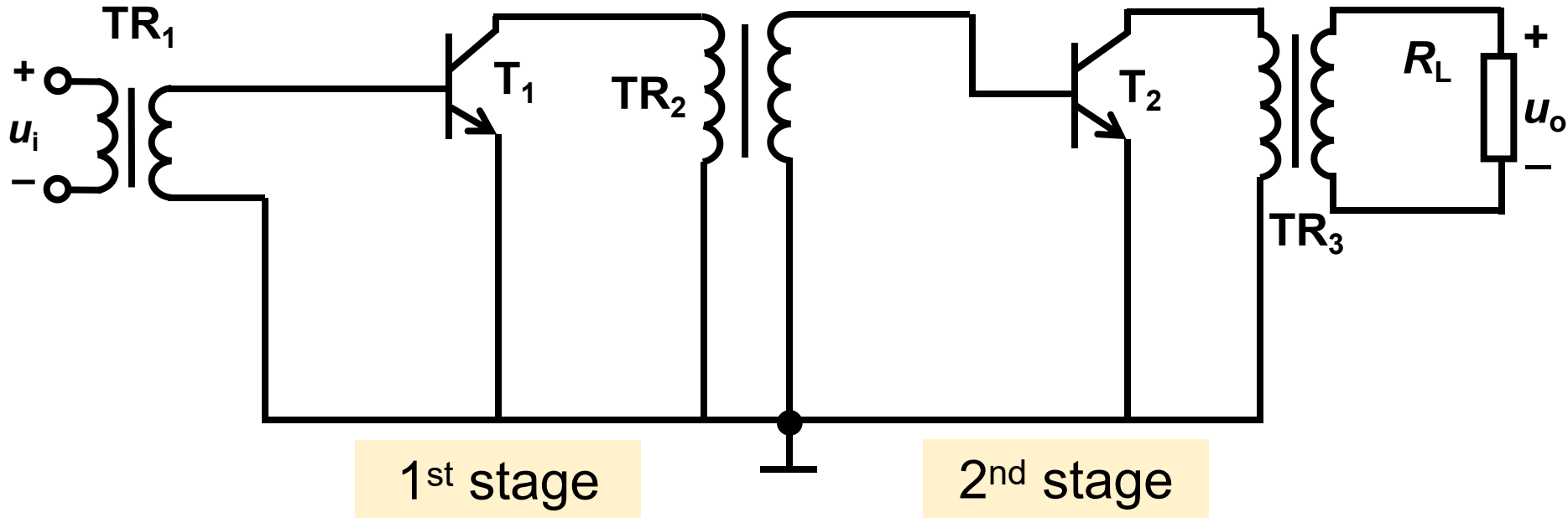


AC circuit

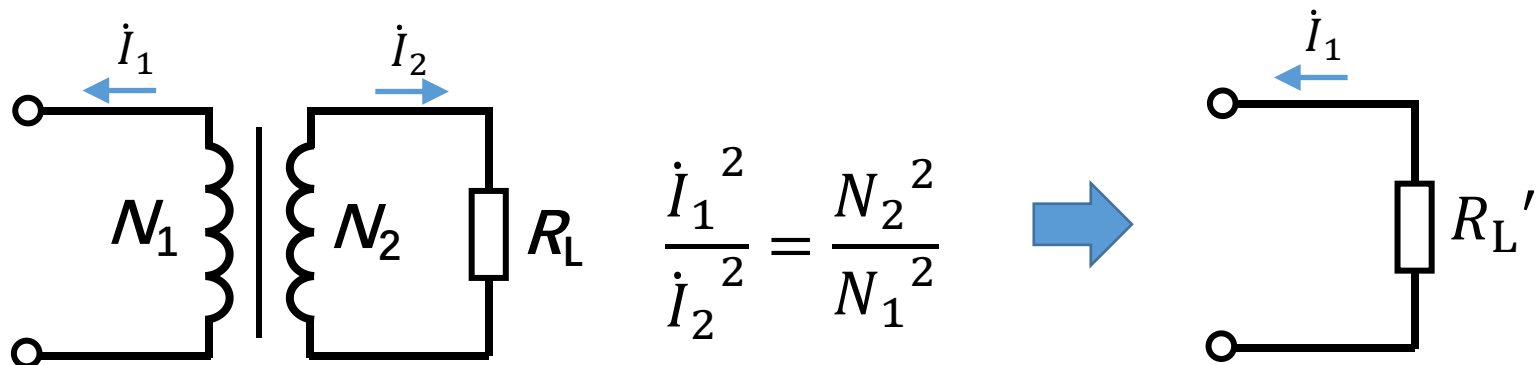


R_{B11} and R_{B21} are short. We can remove them.

AC circuit



Replace transistor with h-model, do it by yourself.



Transformer

Conservation of energy: $i_1^2 R_L' = i_2^2 R_L$

$$R_L' = \frac{N_1^2}{N_2^2} R_L$$

Use transformer, we can change the load resistance and improve the voltage gain

□ Merits of transformer coupling amplification circuit:

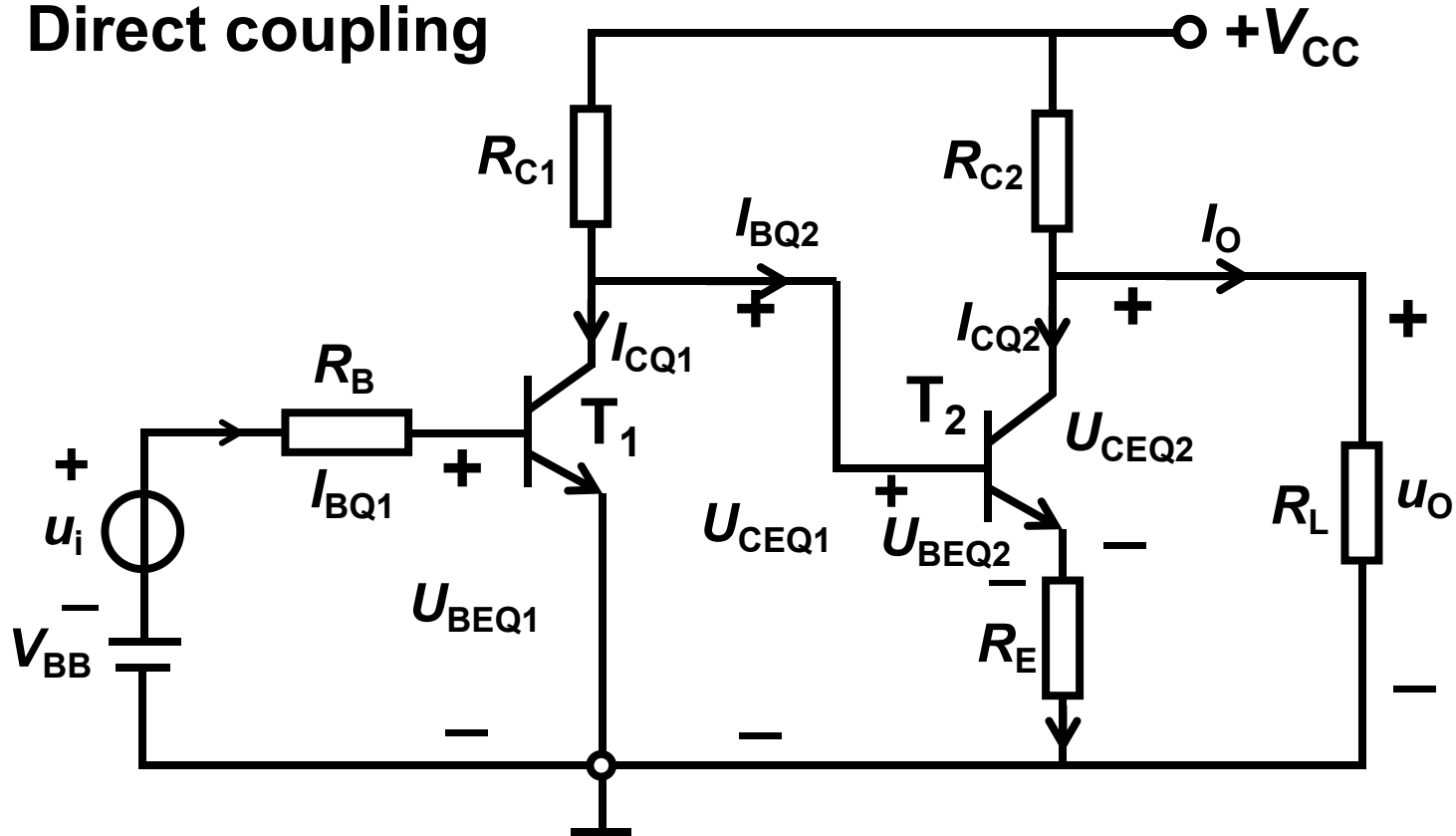
- a) The quiescent working points of every stage are independent**
- b) Can change load resistance**

□ Drawback of transformer coupling amplification circuit:

- a) Cannot amplify DC and low frequency signal**
- b) Cannot be applied to integrated circuits: large volume, expensive**

4.1.3 Analyze multistage amplification circuit

Direct coupling



Analyze DC circuit

We need to get the quiescent working parameters of each stage

Input circuit of 1st stage

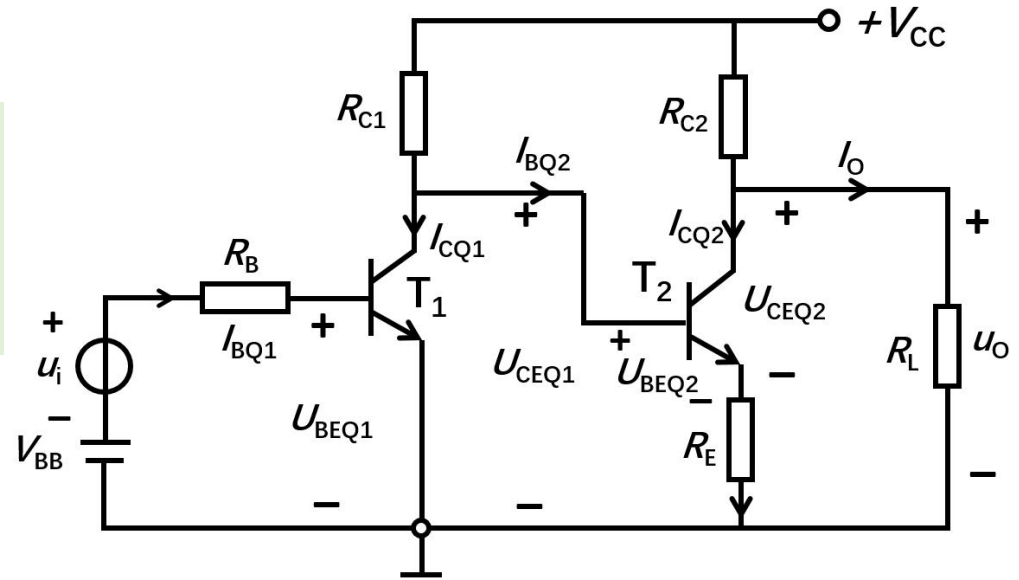
$$I_{BQ1} = \frac{V_{BB} - U_{BEQ1}}{R_B} \quad I_{CQ1} = \bar{\beta}_1 I_{BQ1}$$

Between 1st and 2nd stage

$$U_{CEQ1} = V_{CC} - (I_{CQ1} + I_{BQ2})R_{C1}$$

$$\begin{aligned} U_{CEQ1} &= U_{BEQ2} + I_{EQ2}R_E \\ &= U_{BEQ2} + (1 + \bar{\beta}_2)I_{BQ2}R_E \end{aligned}$$

$$I_{BQ2} = \frac{V_{CC} - U_{BEQ2} - I_{CQ1}R_{C1}}{R_{C1} + (1 + \bar{\beta}_2)R_E}$$



$$I_{CQ2} = \bar{\beta}_2 I_{BQ2}$$

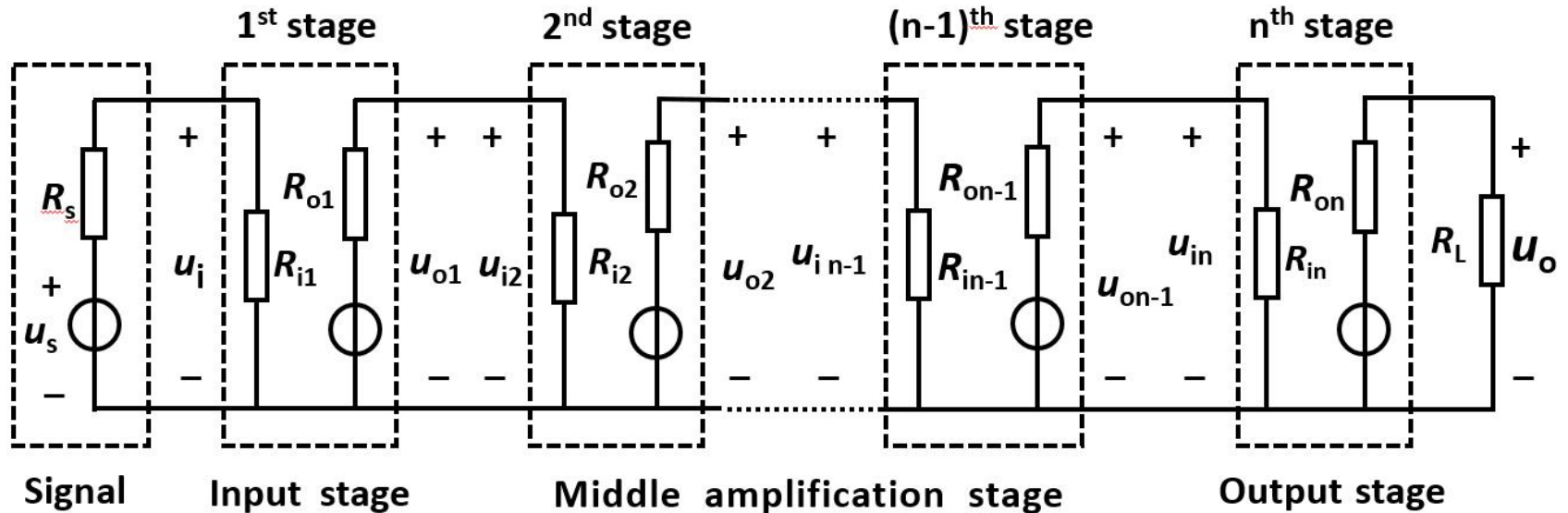
$$I_{EQ2} = (1 + \bar{\beta}_2)I_{BQ2}$$

Output of 2nd stage:

$$V_{CC} = (I_{CQ2} + I_O)R_{C2} + I_O R_L$$

$$U_{CEQ2} = I_O R_L - I_{EQ2} R_E$$

Dynamic analysis:

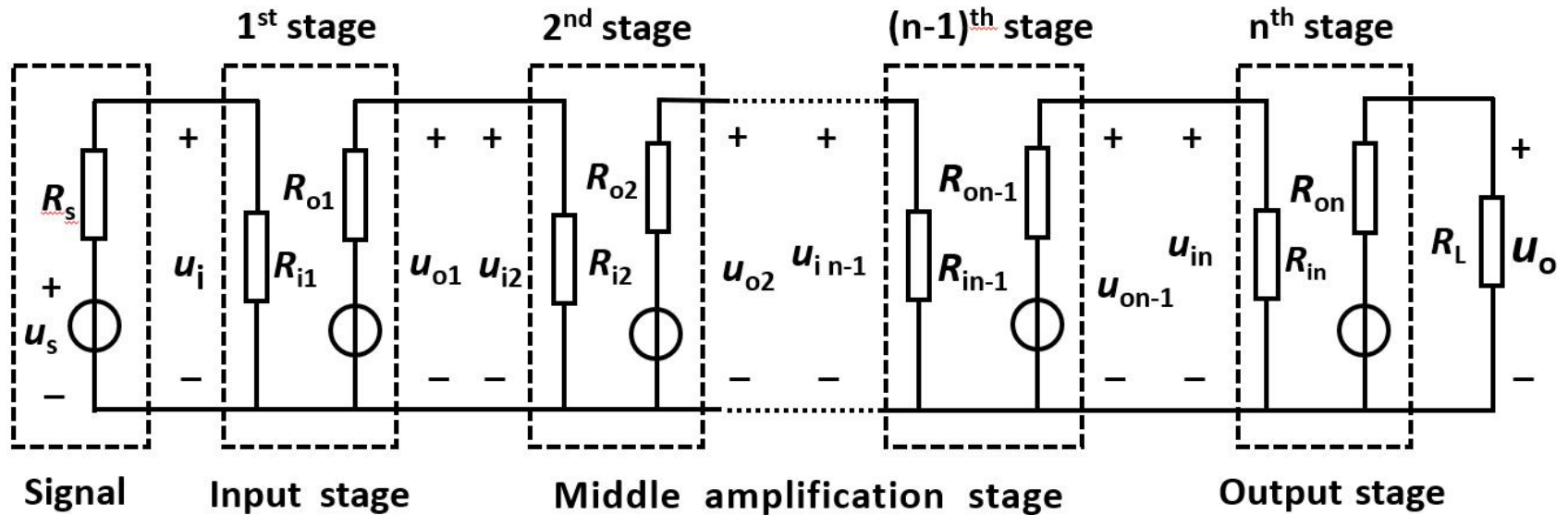


Voltage gain $\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i} = \frac{\dot{U}_{o1}}{\dot{U}_i} \frac{\dot{U}_{o2}}{\dot{U}_{o1}} \frac{\dot{U}_{o3}}{\dot{U}_{o2}} \cdots \frac{\dot{U}_o}{\dot{U}_{on-1}}$

$$\dot{U}_{o1} = \dot{U}_{i2} \quad \dot{U}_{o2} = \dot{U}_{i3} \cdots \quad \dot{U}_{on-1} = \dot{U}_{in}$$

$$\dot{A}_u = \frac{\dot{U}_{o1}}{\dot{U}_i} \frac{\dot{U}_{o2}}{\dot{U}_{i2}} \frac{\dot{U}_{o3}}{\dot{U}_{i3}} \cdots \frac{\dot{U}_{on}}{\dot{U}_{in}}$$

$$\dot{A}_u = \dot{A}_{u1} \dot{A}_{u2} \dot{A}_{u3} \cdots \dot{A}_{un}$$



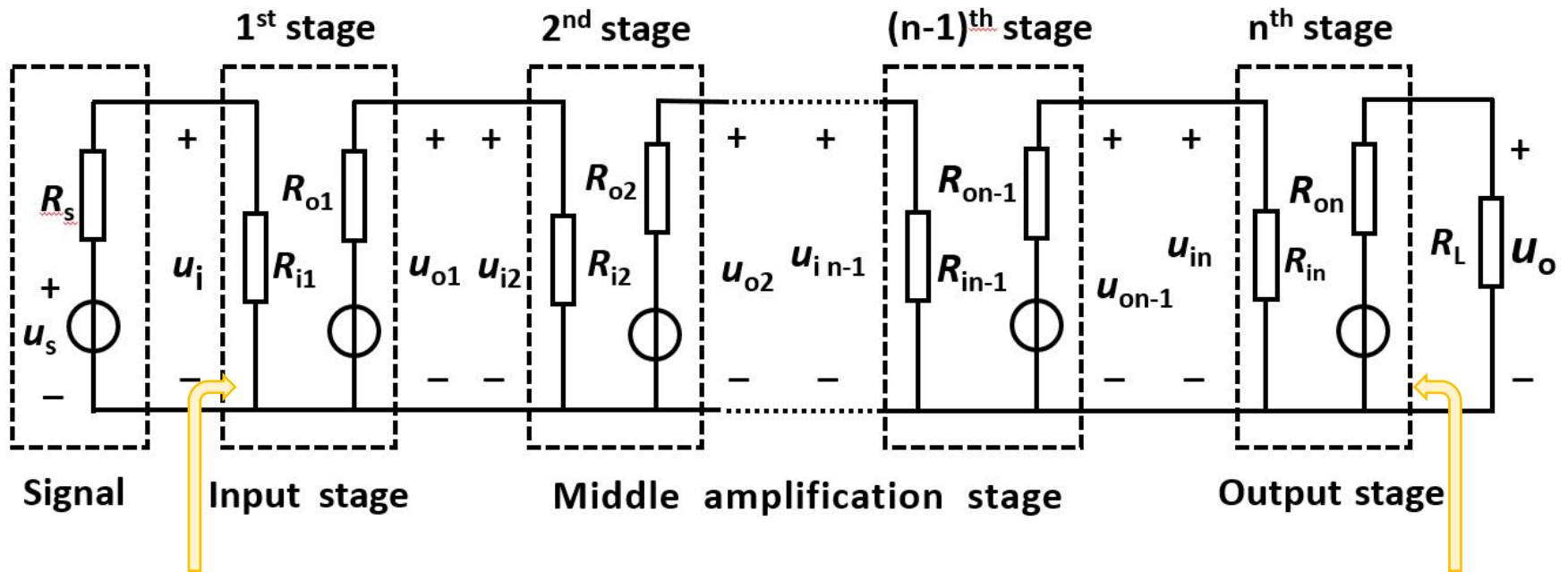
Voltage gain

$$\dot{A}_u = \dot{A}_{u1} \dot{A}_{u2} \dot{A}_{u3} \dots \dot{A}_{un}$$

Introduce another unit: dB (short for decibel)

$$A(\text{dB}) = 20 \log_{10} A = 20 \log A$$

$$\dot{A}_u(\text{dB}) = 20 \log \dot{A}_{u1} + 20 \log \dot{A}_{u2} + 20 \log \dot{A}_{u3} + \dots + 20 \log \dot{A}_{un}$$

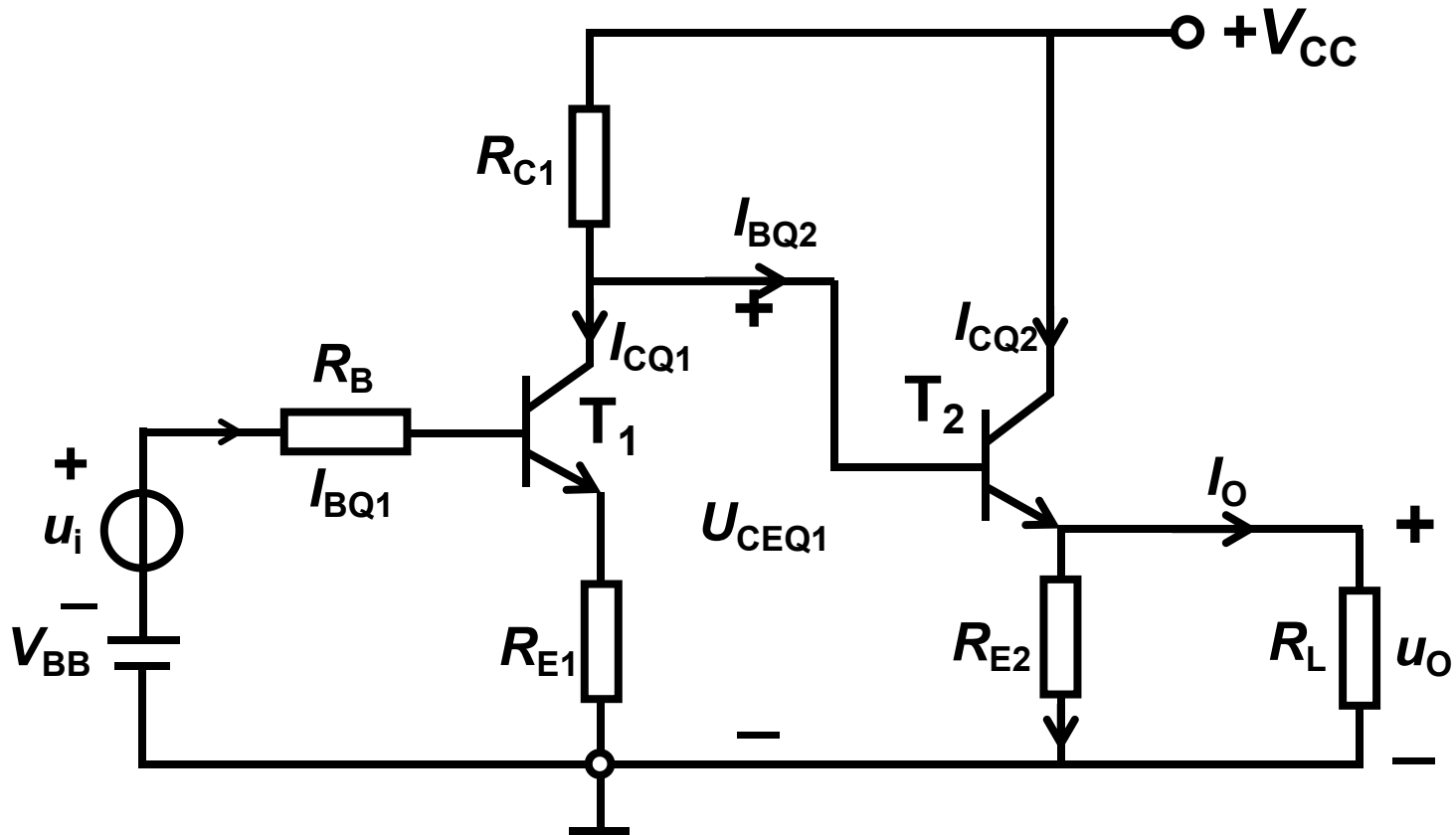


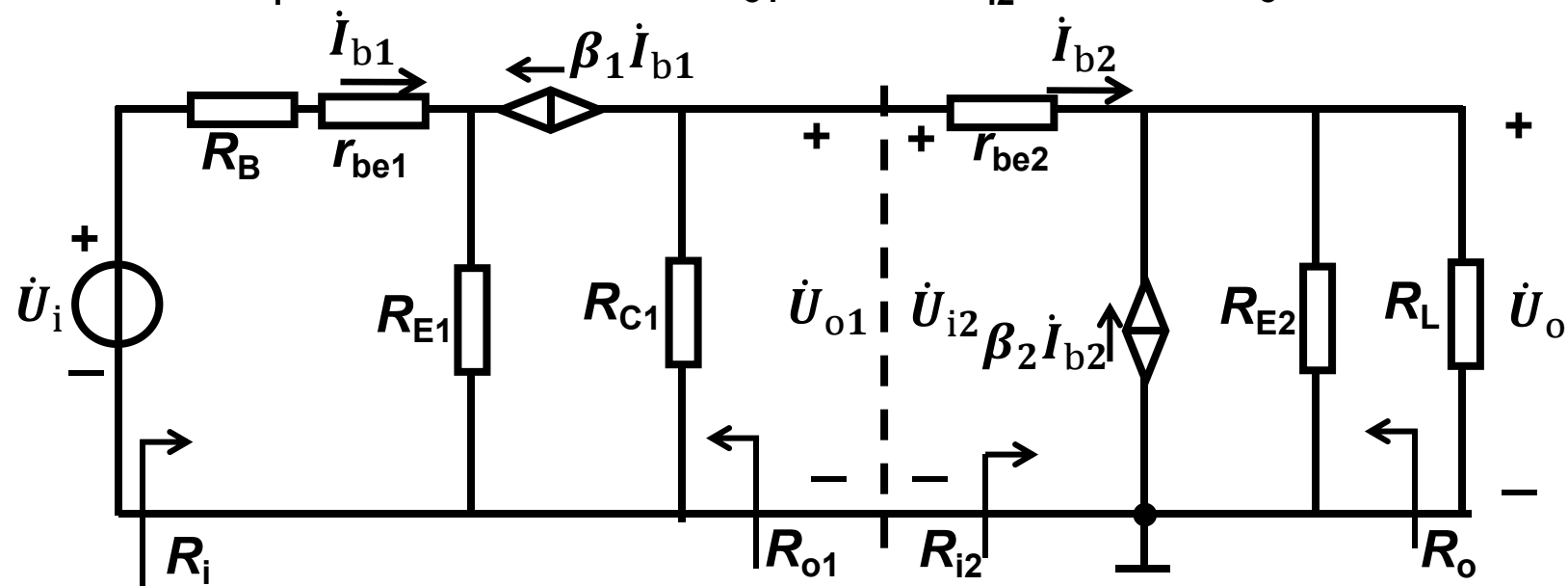
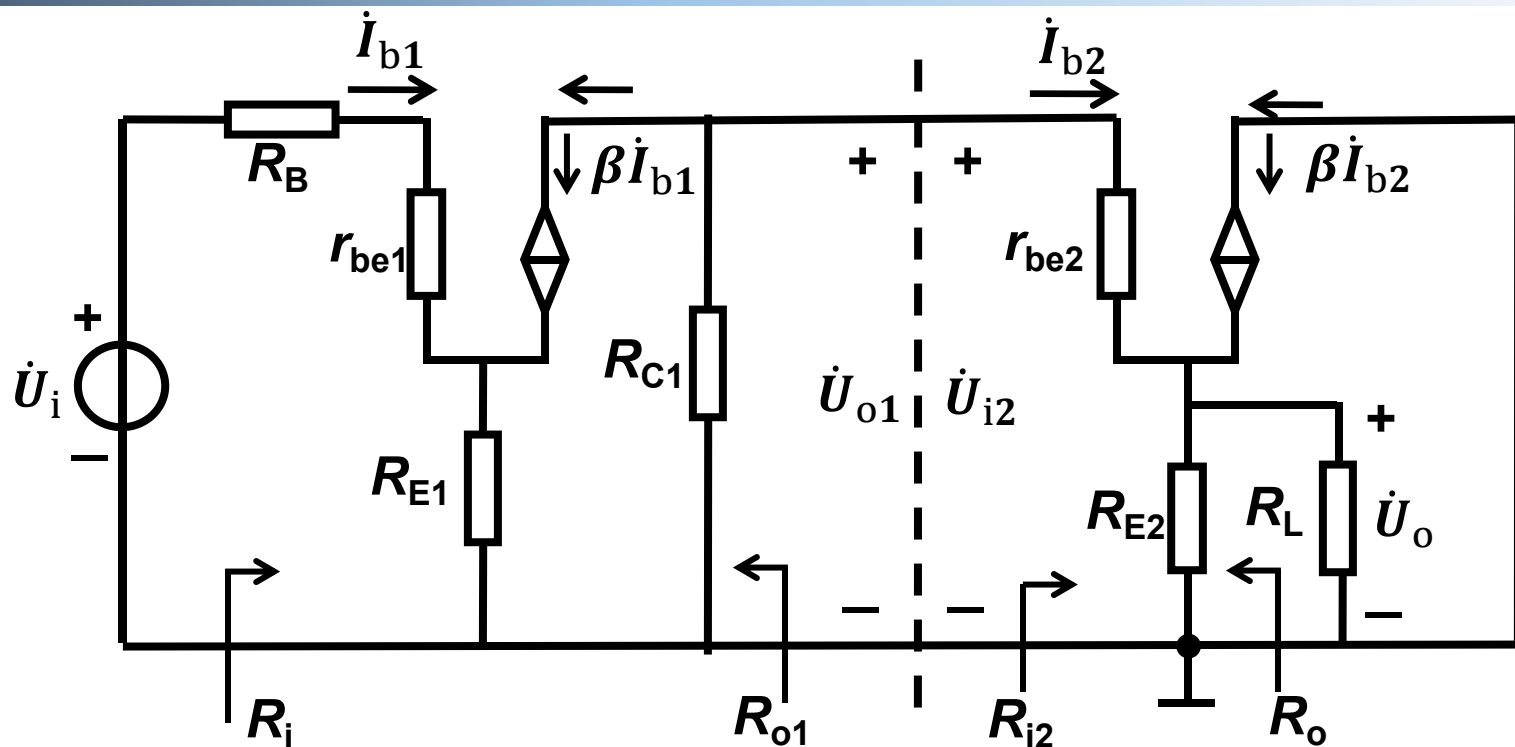
Input resistance R_i

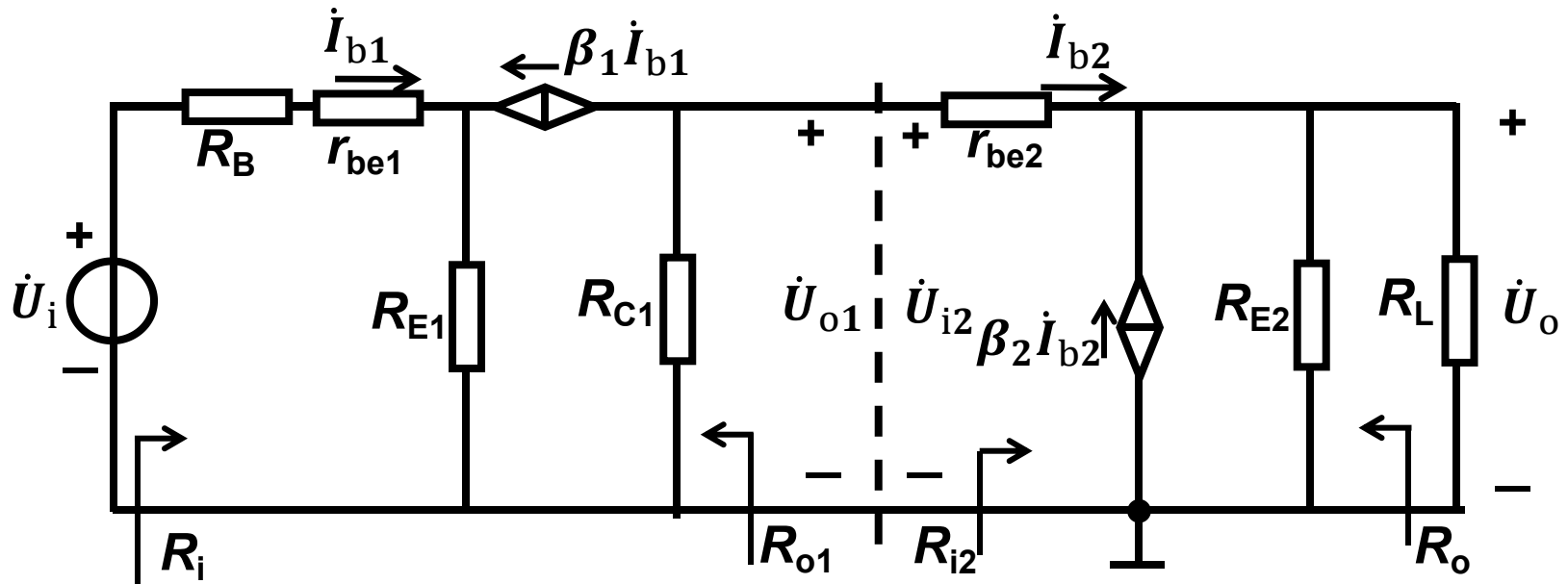
Output resistance R_o

- ❑ The input resistance of $n-1$ stage can be regarded as the load resistance of $n-2$ stage
- ❑ The output resistance of $n-1$ stage can be regarded as the internal resistance of signal source of n stage

Dynamic analysis of the circuit



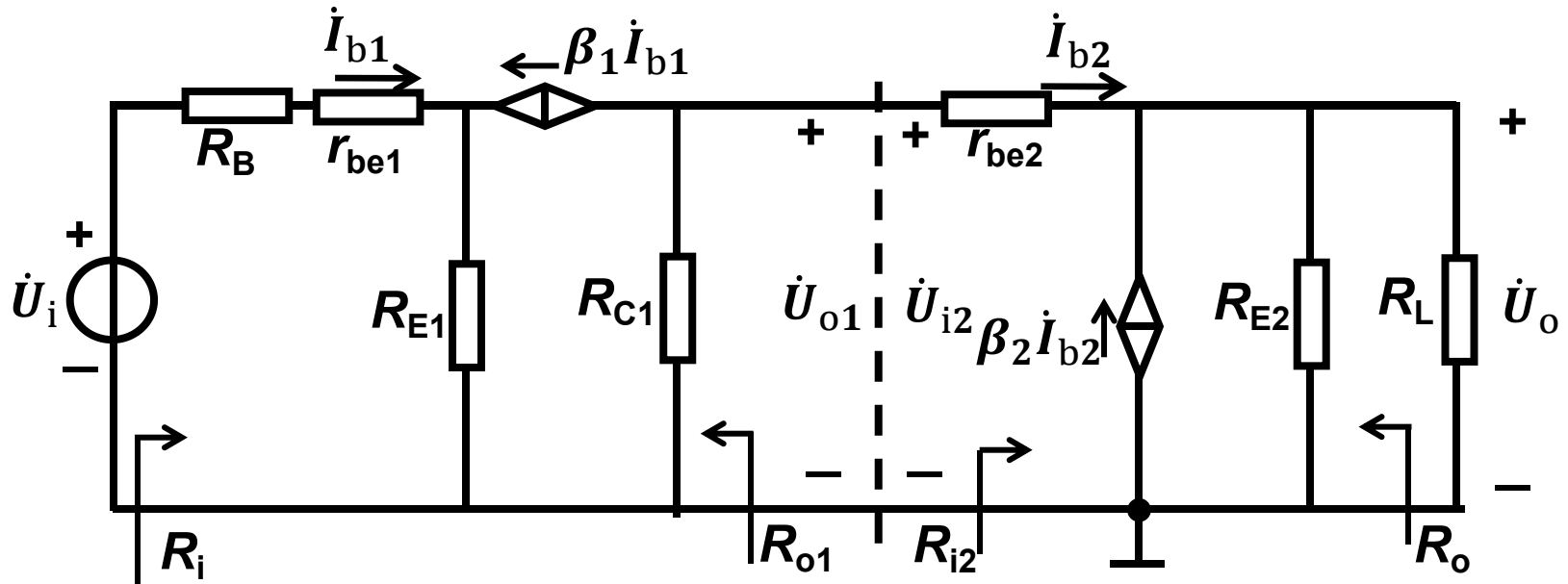




Voltage gain: $\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i}$

1st approach: the violent approach 暴力求解

Try by yourself after class



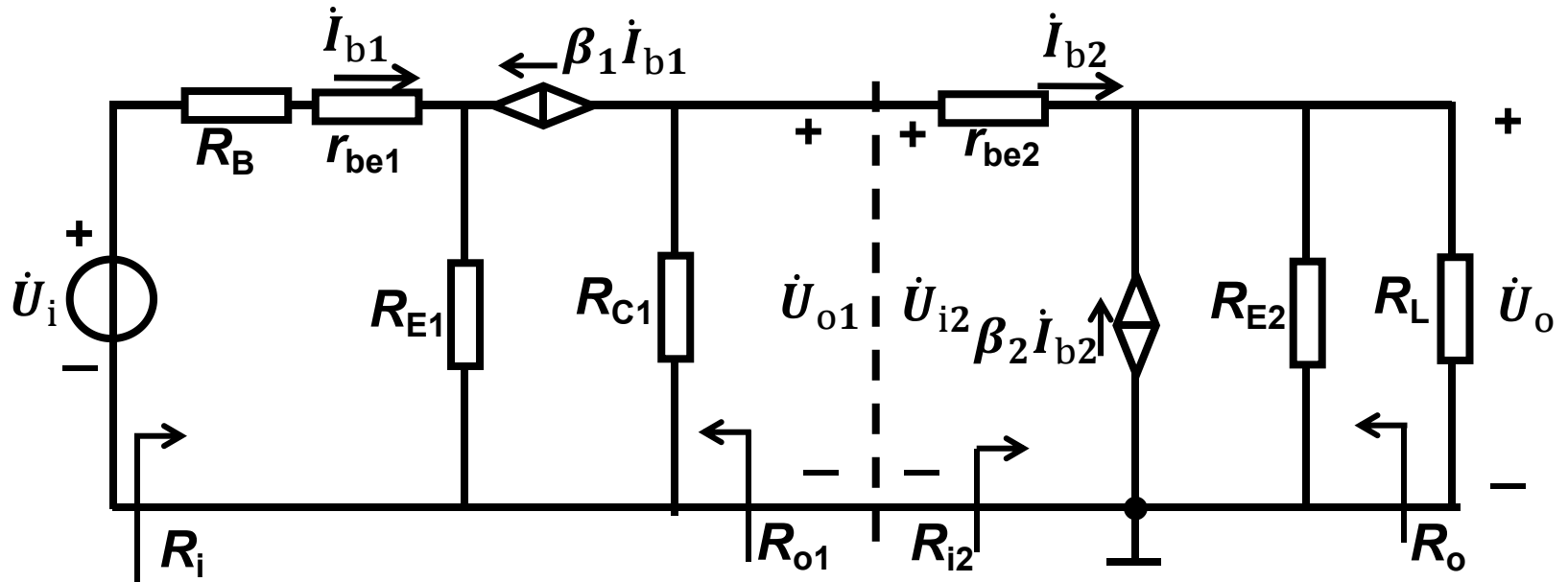
2nd approach

Voltage gain: $\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i}$ $\dot{A}_u = \dot{A}_{u1} \dot{A}_{u2}$

$$\dot{A}_{u1} = \frac{\dot{U}_{o1}}{\dot{U}_i} \begin{cases} \dot{U}_{o1} = -\beta_1 \dot{I}_{b1} (R_{C1} // \textcolor{red}{R}_{i2}) \\ \dot{U}_i = \dot{I}_{b1} (R_B + r_{be1}) + (1 + \beta_1) \dot{I}_{b1} R_{E1} \end{cases}$$

$$\dot{A}_{u1} = \frac{\dot{U}_{o1}}{\dot{U}_i} = - \frac{\beta_1 (R_{C1} // \textcolor{red}{R}_{i2})}{R_B + r_{be1} + (1 + \beta_2) R_{E1}}$$

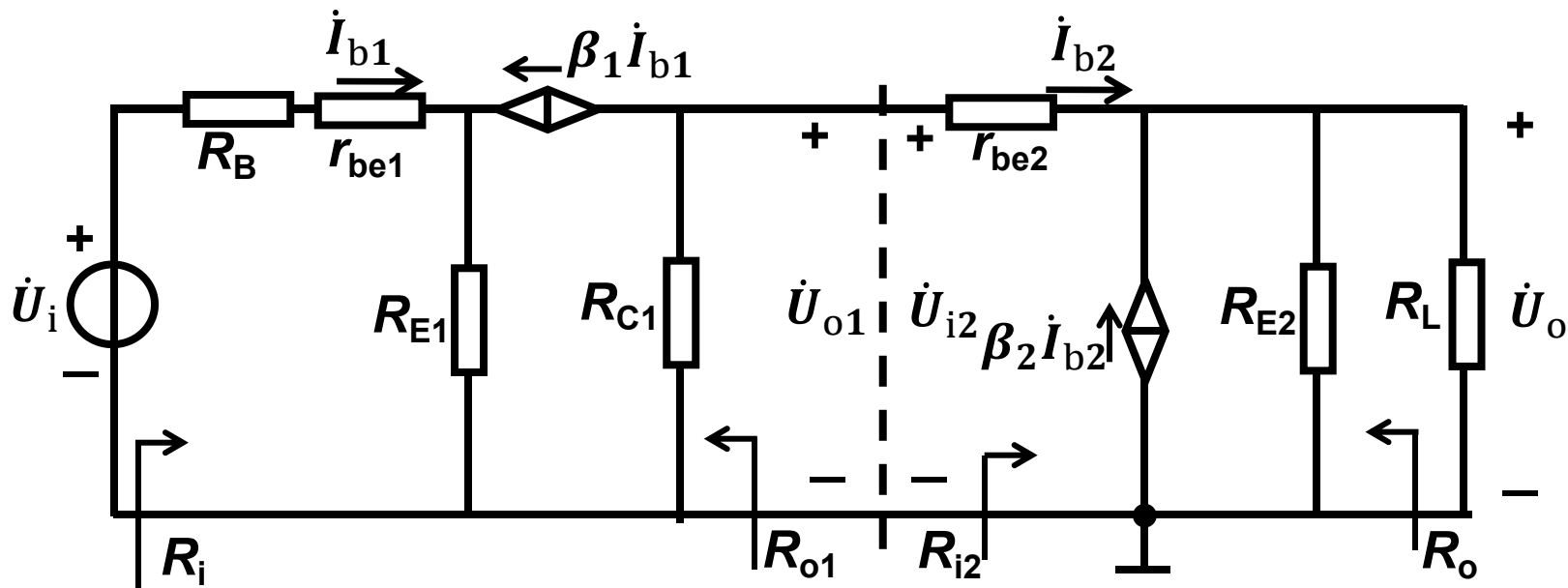
$$\textcolor{red}{R}_{i2} = \frac{\dot{U}_{i2}}{\dot{I}_{b2}} = r_{be2} + (1 + \beta_2) (R_{E2} // R_L)$$



$$\dot{U}_o = (1 + \beta_2) \dot{I}_{b2} (R_L // R_{E2}) \quad \dot{U}_{i2} = \dot{I}_{b2} r_{be2} + \dot{U}_o$$

$$\dot{A}_{u2} = \frac{\dot{U}_o}{\dot{U}_{i2}} = \frac{(1 + \beta_2) (R_L // R_{E2})}{(1 + \beta_2) (R_L // R_{E2}) + r_{be2}}$$

$$\dot{A}_u = \dot{A}_{u1} \dot{A}_{u2} = \frac{\beta_1 (R_{C1} // R_{i2})}{R_B + r_{be1} + (1 + \beta_1) R_{E1}} \frac{(1 + \beta_2) (R_L // R_{E2})}{(1 + \beta_2) (R_L // R_{E2}) + r_{be2}}$$

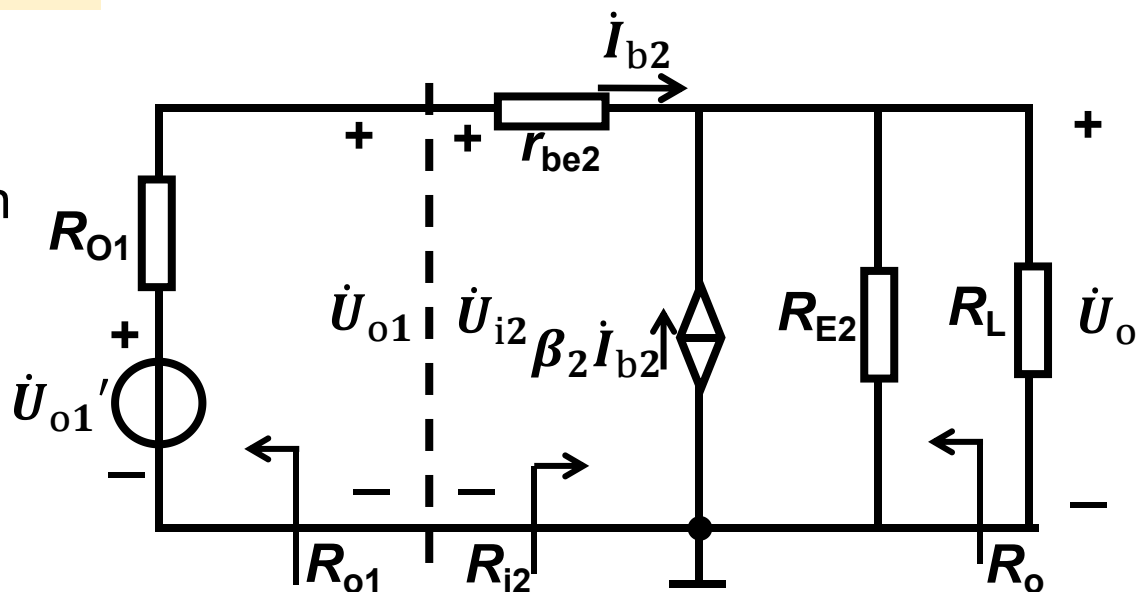


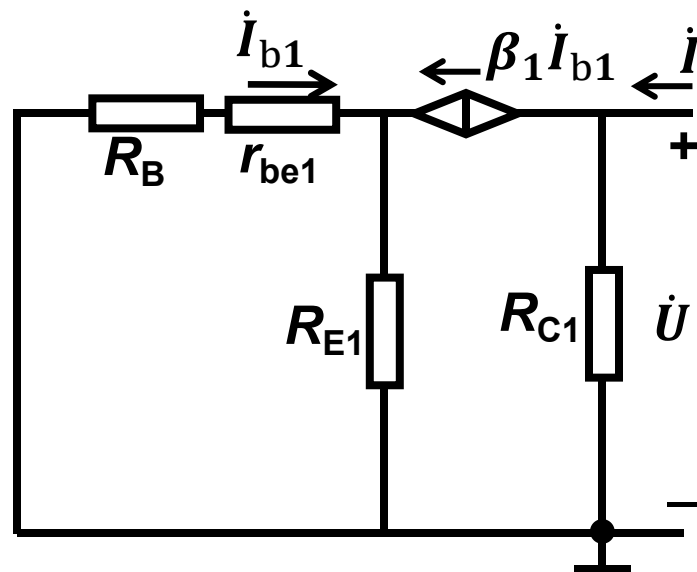
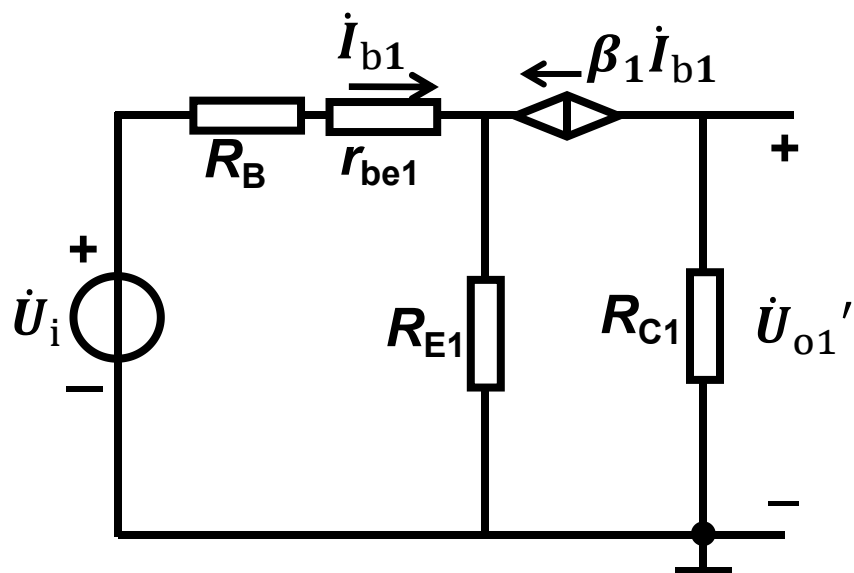
3rd approach

Thevenin's theory 戴维南定理

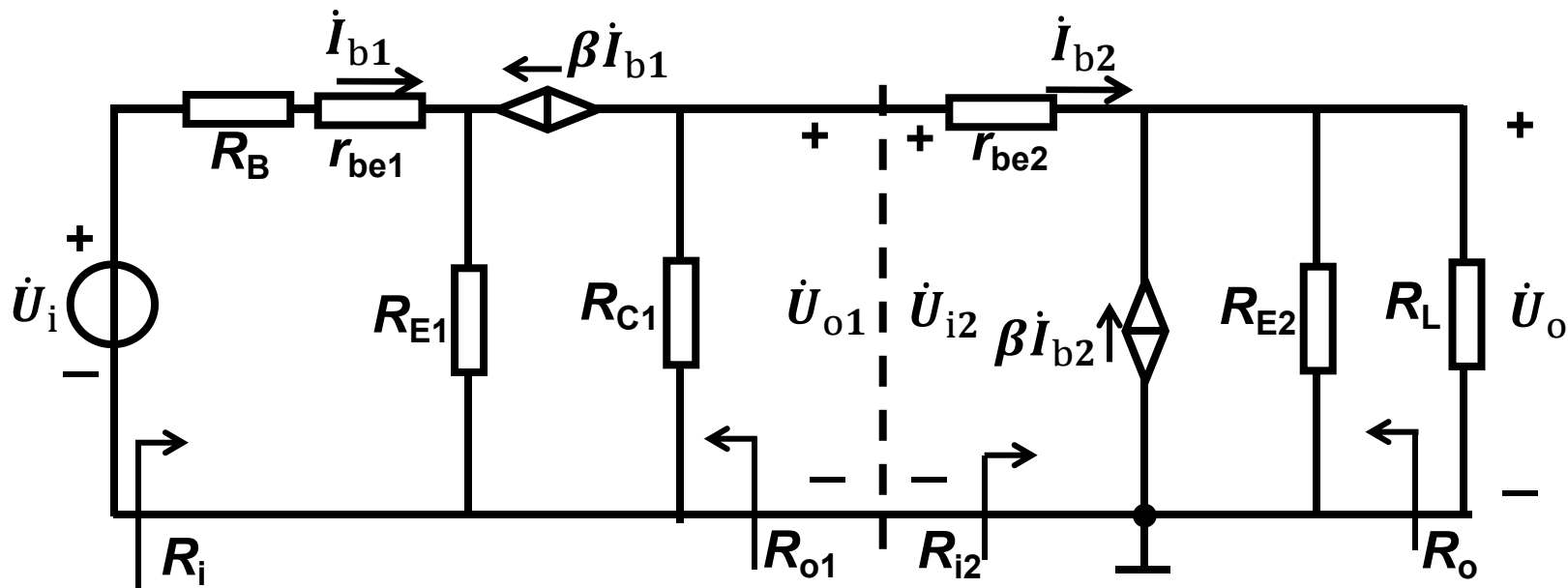
$\dot{U}_{o1}$: the output voltage with load resistor (2nd stage)

$\dot{U}_{o1}'$: the output voltage without load resistor





$$R_{o1} = \frac{\dot{U}}{\dot{I}} \Big|_{R_L=\infty, \dot{U}_i=0}$$

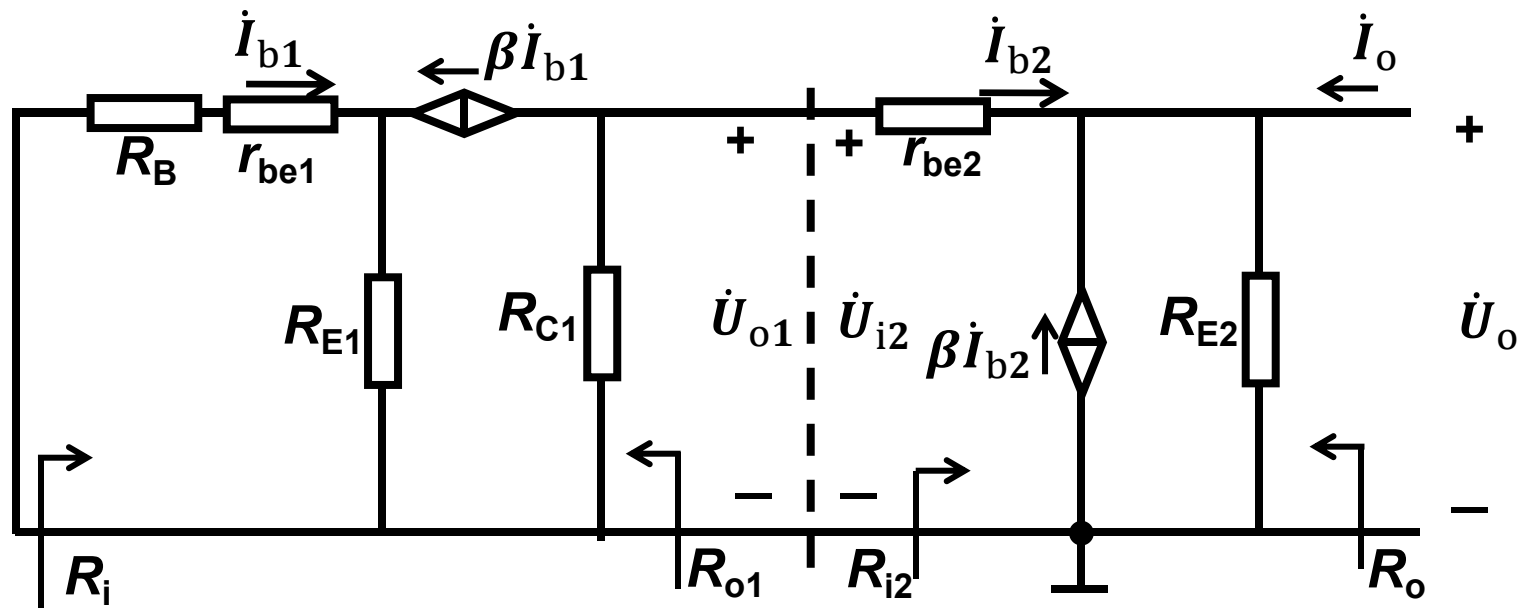


Input resistance: $R_i = \frac{\dot{U}_i}{\dot{I}_i}$

$$\dot{U}_i = \dot{I}_{b1}(R_B + r_{be1}) + (1 + \beta_1)\dot{I}_{b1}R_{E1}$$

$$\dot{I}_i = \dot{I}_{b1}$$

$$R_i = R_B + r_{be1} + (1 + \beta_1) R_{E1}$$



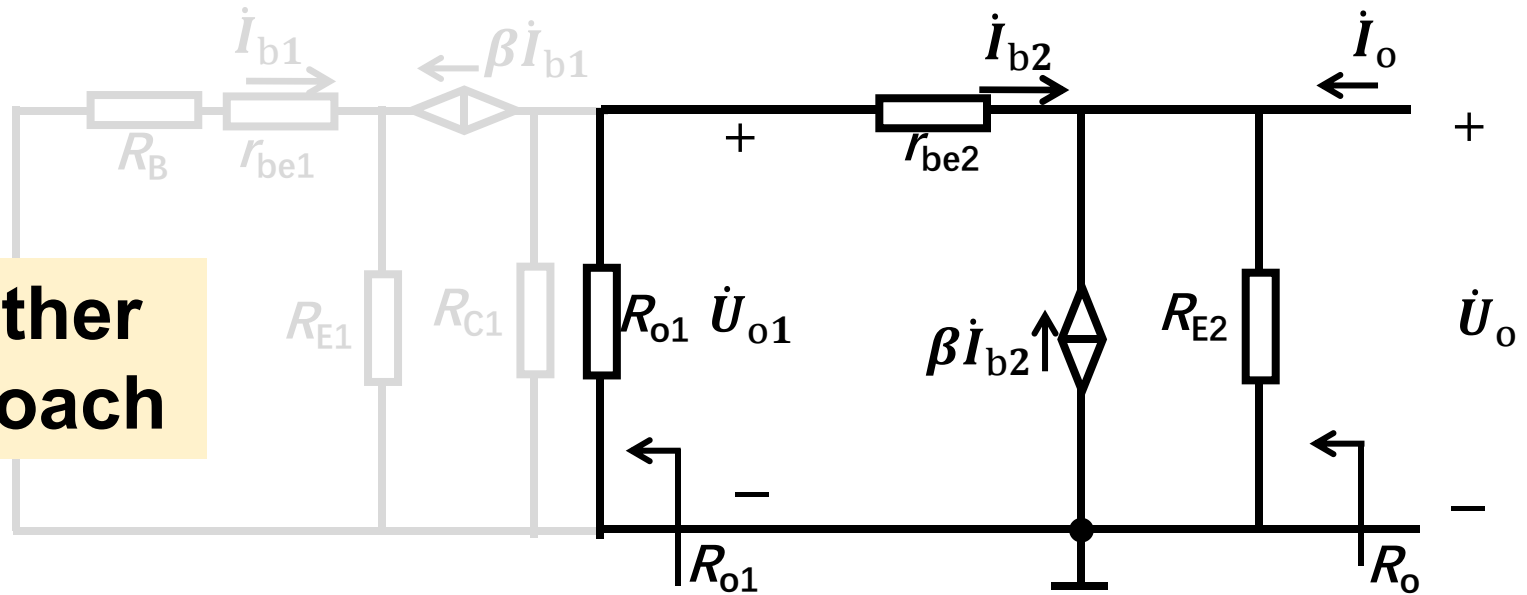
Output resistance: $R_o = \frac{\dot{U}_o}{\dot{I}_o}$

$$\dot{I}_{b1}(R_B + r_{be1}) + (1 + \beta_1)\dot{I}_{b1}R_{E1} = 0 \quad \Rightarrow \quad \dot{I}_{b1} = 0$$

$$\begin{cases} \dot{U}_o = (\dot{I}_{b2} + \beta_2\dot{I}_{b2} + \dot{I}_o)R_{E2} \\ \dot{U}_o = -\dot{I}_{b2}r_{be2} - \dot{I}_{b2}R_{C1} \end{cases} \quad \Rightarrow \quad \dot{I}_o = -\frac{r_{be2} + R_{C1} + (1 + \beta_2)R_{E2}}{R_{E2}}\dot{I}_{b2}$$

$$\Rightarrow R_o = R_{E2} // \frac{r_{be2} + R_{C1}}{1 + \beta_2}$$

Another approach



Output resistance:

$$R_o = R_{E2} // \frac{r_{be2} + R_{O1}}{1 + \beta_2}$$

$$R_{O1} = R_{C1}$$

Homework 3-2:

Please hand in homework 3-1 and 3-2 together

Deadline: 2025-04-09 11:59pm